

# Information Visualisation (COMPSCI5099) -- Exam Notes

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**University of Glasgow | MSc Computing Science Exam:** Thursday 07/05/2026, 14:00 -- 1h 30m | Closed Book, No Calculator | Digital On Campus

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# 1. Introduction to Information Visualisation

## Definitions

- **Information Visualisation (InfoVis):** The use of computer-supported, interactive, visual representations of abstract data to amplify cognition.
- **Scientific Visualisation (SciVis):** Visualisation of physically-based data (3D volumes, flow fields). Data has inherent spatial mapping.
- **InfoVis vs SciVis:** InfoVis deals with *abstract* data (no inherent spatial mapping); the designer must *choose* spatial layout. SciVis data already has a natural spatial representation.

## Why Visualise?

- Humans have **remarkable perceptual abilities** -- we can scan, recognise, and recall images rapidly
- Visual displays exploit the **high bandwidth** of the human visual system
- Visualisation reveals **patterns, clusters, gaps, and outliers** that are invisible in raw data
- Supports **exploration** (finding unexpected patterns) and **explanation** (communicating known findings)

## Key Distinction: Exploration vs Explanation

| Aspect      | Exploratory               | Explanatory                 |
|-------------|---------------------------|-----------------------------|
| Goal        | Discover unknown patterns | Communicate known findings  |
| Audience    | Analyst (self)            | Others (presentation)       |
| Interaction | Highly interactive        | Often static                |
| Design      | Flexible, multiple views  | Polished, focused narrative |

**Exam Tip:** Be able to distinguish InfoVis from SciVis and give examples of each. A scatter plot of GDP vs life expectancy = InfoVis. A 3D MRI brain scan = SciVis.

## 2. Visual Variables -- Bertin's Semiology of Graphics

### Background

**Jacques Bertin (1918-2010)** -- French cartographer who published *Semiologie Graphique* (1967). His framework defines how visual marks encode data and remains foundational to visualisation design.

**Semiotics** is the study of signs and how they convey meaning. Visualisation is a *visual language* with tokens (signs) and syntax (rules for combining them).

### Types of Signs

- **Symbols:** No perceptual relationship to what they represent (arbitrary) -- e.g., a red circle meaning "danger"
- **Icons:** Clear perceptual relationship to what they represent (non-arbitrary) -- e.g., a picture of a house

### Two Categories of Visual Variables

**1. Location (Positional) Variables** - Position relative to a coordinate frame (x, y axes) - Fix a "graphic mark" onto a position on the plane - Most powerful channel for encoding data

**2. Retinal Variables (6 total)** - "Elevate" a mark with a different "pattern of light" - **Size:** measured in mm or pixels - **Shape:** e.g., square, circle, star, triangle - **Colour (Hue):** wavelength -- blue, green, red - **Colour (Value):** depth/lightness of colour (greyscale equivalent); red with low value = pink - **Orientation:** angle of the most prominent axis relative to coordinate axes - **Texture:** spacing between repeated elements (fine vs coarse)

### Four Perceptual Properties of Visual Variables

| Property            | Definition                                           | Which Variables?                           |
|---------------------|------------------------------------------------------|--------------------------------------------|
| <b>Associative</b>  | All variations perceived equally; no value dominates | Location, shape, orientation, hue, texture |
| <b>Dissociative</b> | One variation dominates others; eye drawn to it      | Size, colour value                         |
| <b>Selective</b>    | Can focus on one variation despite changes in others | All EXCEPT shape                           |
| <b>Ordered</b>      | Variations perceived as ranked                       | Size, colour value, texture                |
| <b>Quantitative</b> | Can estimate numerical magnitude of differences      | Location, size                             |

### Matching Variables to Data Types

| Data Type                                 | Best Visual Variables            | Rationale                         |
|-------------------------------------------|----------------------------------|-----------------------------------|
| <b>Nominal</b> (categories, no order)     | Shape, hue, orientation, texture | Associative -- no value dominates |
| <b>Ordinal</b> (ordered, non-numeric)     | Colour value, size               | Dissociative + ordered            |
| <b>Quantitative</b> (numeric, measurable) | Position, size                   | Quantitative perception possible  |

## Bertin's Effectiveness Table (Simplified)

| Variable    | Nominal  | Ordinal  | Quantitative | Selective |
|-------------|----------|----------|--------------|-----------|
| Position    | Good     | Good     | Good         | Yes       |
| Size        | Marginal | Good     | Good         | Yes       |
| Shape       | Good     | Poor     | Poor         | <b>No</b> |
| Value       | Poor     | Good     | Marginal     | Yes       |
| Hue         | Good     | Poor     | Poor         | Yes       |
| Orientation | Marginal | Poor     | Poor         | Yes       |
| Texture     | Good     | Marginal | Poor         | Yes       |

**Exam Tip:** Shape is the ONLY non-selective variable. This is a favourite exam question. Also remember: position and size are the only quantitative variables.

## Figure and Ground

- **Figure:** the variation that stands out most
- **Ground:** the variation that recedes into the background
- Dissociative variables (size, value) create strong figure-ground separation

### 3. Data Types and Data Attributes

#### Munzner's Data Attribute Types

| Type                         | Description                   | Examples                        | Operations        |
|------------------------------|-------------------------------|---------------------------------|-------------------|
| <b>Categorical (Nominal)</b> | No implicit order             | Fruit names, countries          | =, !=             |
| <b>Ordinal</b>               | Implicit order, non-numeric   | Low/Medium/High, T-shirt sizes  | =, !=, <, >       |
| <b>Quantitative</b>          | Numeric, measurable distances | Temperature, salary, population | =, !=, <, >, +, - |

#### Shneiderman's Seven Data Types

1. **1-Dimensional:** Sequential data -- text documents, source code, lists
2. **2-Dimensional:** Planar/map data -- geographic maps, floor plans
3. **3-Dimensional:** Volumetric -- molecules, buildings, human body
4. **Temporal:** Time-varying data -- stock prices, medical records, project timelines
5. **Multi-dimensional:** Items with many attributes -- relational database records
6. **Tree:** Hierarchical data -- file systems, org charts, taxonomies
7. **Network:** Items with arbitrary relationships -- social networks, web links

#### Dataset Types (Munzner)

- **Tables:** rows (items) x columns (attributes)
- **Networks/Graphs:** nodes + links (edges)
- **Trees:** hierarchical, connected, acyclic graph
- **Fields:** continuous data, sampled at grid positions
- **Geometry:** spatial positions and shapes
- **Sets/Clusters:** groupings of items

**Exam Tip:** Know Shneiderman's 7 data types by heart and be able to give an example visualisation for each.

## 4. The Visualisation Pipeline

### The Pipeline Model

Raw Data --> Data Transformation --> Visual Mapping --> View Transformation --> Rendering -

**Stage 1: Data Transformation** - Filtering, cleaning, aggregating, deriving new attributes - Converting data into a form suitable for visualisation

**Stage 2: Visual Mapping** (most critical stage) - Map data attributes to visual variables - Choose marks (points, lines, areas) and channels (position, colour, size) - This is where the designer's decisions most affect effectiveness

**Stage 3: View Transformation** - Camera/viewpoint changes, zooming, panning - Focus+context techniques - Navigation and orientation

**Stage 4: Rendering** - Generate the final visual output (pixels on screen)

### Marks and Channels (Munzner)

**Marks** = geometric primitives representing data items: - **Points** (0D) -- individual items - **Lines** (1D) -- connections/relationships - **Areas** (2D) -- regions/containment - **Volumes** (3D) -- 3D regions

**Channels** = visual properties that encode data attributes: - **Magnitude channels** (ordered): position, length, angle, area, volume, luminance, saturation - **Identity channels** (categorical): spatial region, colour hue, motion, shape

### Channel Effectiveness Ranking (Munzner, after Cleveland & McGill)

**For quantitative data (most to least effective):** 1. Position on a common scale 2. Position on unaligned scales 3. Length 4. Angle / Slope 5. Area 6. Volume 7. Colour luminance / Saturation

**For categorical data (most to least effective):** 1. Spatial region 2. Colour hue 3. Motion 4. Shape

**Exam Tip:** Position on a common scale is ALWAYS the most effective channel for quantitative data. This is why bar charts and scatter plots work so well.

## 5. Interaction Techniques -- Shneiderman's Taxonomy

### The Visual Information-Seeking Mantra

"Overview first, zoom and filter, then details-on-demand" -- Ben Shneiderman, "The Eyes Have It" (1996)

This is the single most important principle in interactive information visualisation.

### Shneiderman's Seven Tasks

| Task              | Description                                              | Example                              |
|-------------------|----------------------------------------------------------|--------------------------------------|
| Overview          | Gain overview of the entire collection                   | See all 10,000 data points at once   |
| Zoom              | Zoom in on items of interest                             | Magnify a cluster in a scatter plot  |
| Filter            | Filter out uninteresting items                           | Show only records from 2024          |
| Details-on-demand | Select an item, get details when needed                  | Tooltip on hover showing full record |
| Relate            | View relationships among items                           | Highlight connections in a network   |
| History           | Keep history for undo, replay, refinement                | Step back through exploration states |
| Extract           | Allow extraction of sub-collections and query parameters | Export filtered subset as CSV        |

### Task by Data Type Taxonomy (TTT)

Shneiderman proposes a matrix: **7 data types x 7 tasks**. Each cell represents a specific design challenge. The taxonomy helps designers identify which combinations have been explored and which remain open.

### Interaction Techniques in Practice

**Overview + Detail:** - Show both overview and detailed view simultaneously - Example: Google Maps -- minimap + main view

**Focus + Context:** - Show focus area in detail while keeping surrounding context visible - Example: Fisheye lens distortion, bifocal display

**Brushing and Linking:** - Select (brush) items in one view; corresponding items are highlighted in all linked views - Essential for coordinated multiple views

**Dynamic Queries:** - Use sliders, buttons, toggles to filter data in real-time - Immediate visual feedback; encourages exploration - Example: Homefinder (Shneiderman) -- sliders for price, bedrooms, distance

**Direct Manipulation:** - Objects of interest are visible - Actions are rapid, incremental, reversible - Feedback is immediate and continuous

**Exam Tip:** The mantra "Overview first, zoom and filter, then details-on-demand" could appear in almost any question. Be able to apply it to a given scenario.

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## 6. Data Selection and Filtering

### Heer et al. -- Generalized Selection via Interactive Query Relaxation

**Core Idea:** Traditional selection (click/lasso individual items) is too limited. Users need *declarative* selections that express higher-level criteria.

### Key Concepts

- **Selection as Query:** Model selection as a declarative query (SQL-like) over data attributes, not just a set of selected items
- **Query Relaxation:** Users start with a specific selection and *generalise* it interactively -- "select all items like this one"
- **Advantages:** Supports both static and dynamic interface objects; selections persist as data updates

### Types of Selection

| Type                         | Description                      | Example                        |
|------------------------------|----------------------------------|--------------------------------|
| <b>Point selection</b>       | Click on individual items        | Select one data point          |
| <b>Region selection</b>      | Lasso/rubber-band a spatial area | Select all items in East L.A.  |
| <b>Attribute selection</b>   | Select by data attributes        | "All gunshot victims"          |
| <b>Generalised selection</b> | Relax from specific to general   | "All items blue like this one" |

### Selection Operations

- **Union:** Combine two selections (OR)
- **Intersection:** Items in both selections (AND)
- **Difference:** Items in first but not second
- **Complement:** Everything NOT selected

### Dynamic Queries vs Query Relaxation

| Dynamic Queries             | Query Relaxation            |
|-----------------------------|-----------------------------|
| Start broad, narrow down    | Start specific, broaden out |
| Slider-based filtering      | Click-and-generalise        |
| Top-down approach           | Bottom-up approach          |
| Predefined query dimensions | Emergent query dimensions   |

**Exam Tip:** Heer's key contribution is coupling declarative query structure with interactive generalisation. Users create more accurate selections with generalisation than without.

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## 7. Visual Perception and Pre-attentive Processing

### Pre-attentive Processing

**Definition:** Visual features that are detected by the low-level visual system *before* conscious attention -- typically in under 200-250 milliseconds, regardless of the number of distractors.

- Recognised "pre-conceptually" at a **sensory level** (not cognitive)
- "Seen" rather than "understood"
- Often called "**pop-out**" (Treisman, 1985)

### Pre-attentive Features

The following visual properties support pre-attentive processing: - **Colour (hue and value)** - **Size** (length, width, area) - **Orientation** (angle) - **Shape** (basic forms -- but with limits) - **Motion** (flicker, direction) - **Spatial position** - **Curvature** - **Added marks** (e.g., enclosure)

### Treisman's Feature Integration Theory

1. **Pre-attentive stage:** Individual features (colour, orientation) processed in parallel across the visual field
2. **Attentive stage:** Features are combined ("bound") through focused attention -- serial, slower
3. **Conjunction searches** (find the red circle among red squares and blue circles) are NOT pre-attentive; they require serial scanning

### Gestalt Principles of Perception

| Principle            | Description                                     | Use in Visualisation       |
|----------------------|-------------------------------------------------|----------------------------|
| <b>Proximity</b>     | Items close together perceived as grouped       | Spacing in small multiples |
| <b>Similarity</b>    | Items that look alike perceived as grouped      | Colour coding categories   |
| <b>Continuity</b>    | Eye follows smooth paths                        | Line charts, flow lines    |
| <b>Closure</b>       | Mind completes incomplete shapes                | Implied boundaries         |
| <b>Connectedness</b> | Connected items perceived as grouped            | Links in network diagrams  |
| <b>Enclosure</b>     | Items within a boundary perceived as grouped    | Bounding boxes, regions    |
| <b>Common fate</b>   | Items moving together perceived as grouped      | Animated transitions       |
| <b>Figure-ground</b> | Perception separates foreground from background | Contrast, layering         |

### Weber's Law and Just Noticeable Difference (JND)

- **Weber's Law:** The just noticeable difference between two stimuli is proportional to the magnitude of the stimuli
- $\Delta S / S = k$  (where  $S$  = stimulus magnitude,  $k$  = constant)
- Implications: Small differences are easier to detect at small magnitudes; harder at large magnitudes
- This is why **position on a common scale** is the most effective channel -- the JND is smallest

## Stevens' Power Law

- Perceived magnitude  $P = k * S^n$  (S = physical stimulus, n = exponent)
- Different visual channels have different exponents:
- **Length:**  $n \sim 1.0$  (perceived proportionally)
- **Area:**  $n \sim 0.7$  (underestimated)
- **Volume:**  $n \sim 0.6$  (even more underestimated)
- **Brightness:**  $n \sim 0.5$

**Exam Tip:** Pre-attentive processing is a core concept. Be ready to explain why a red dot pops out among blue dots, but finding a red circle among red squares and blue circles does not.

## 8. Colour Theory

### Colour Spaces

**RGB (Red, Green, Blue)** - Additive colour model (light-based) - Used in displays/screens - Not perceptually uniform -- equal steps in RGB do not look like equal steps to the eye

**HSL / HSV (Hue, Saturation, Lightness/Value)** - More intuitive for design - Hue = colour wheel position (0-360 degrees) - Saturation = purity/intensity of colour - Lightness/Value = brightness

**CIELAB (Lab\*)** - Designed to be **perceptually uniform** -- equal numerical changes correspond to equal perceived changes - L = *lightness* (0-100), a = green-red, b\* = blue-yellow - Preferred for scientific visualisation and colour map design

### Colour in Visualisation

**Hue** (categorical/nominal data): - Different hues = different categories - Associative: no hue perceived as dominant - Limit to 5-8 distinguishable hues (more is confusing)

**Value/Luminance** (ordinal/quantitative data): - Dark-to-light encodes low-to-high - Ordered and dissociative - Works for sequential data

**Saturation** (limited use): - Low saturation = grey/muted; high saturation = vivid - Less effective than luminance for ordered data - Can encode a secondary variable

### Colour Map Types

| Type                           | Description                      | Use Case                    | Example                                |
|--------------------------------|----------------------------------|-----------------------------|----------------------------------------|
| <b>Sequential</b>              | Single hue, varying lightness    | Ordered data (low to high)  | Light yellow to dark blue              |
| <b>Diverging</b>               | Two hues from a neutral midpoint | Data with meaningful centre | Blue-white-red for temperature anomaly |
| <b>Categorical/Qualitative</b> | Distinct hues, similar lightness | Nominal categories          | Category A=blue, B=orange, C=green     |

### Colour Blindness

- ~8% of males, ~0.5% of females have colour vision deficiency
- **Protanopia/Deutanopia**: Cannot distinguish red from green (most common)
- **Tritanopia**: Cannot distinguish blue from yellow (rare)

**Design Guidelines**: - Never use red-green as the only differentiator - Use **redundant encoding** (colour + shape, colour + texture) - Use colour-blind-safe palettes (e.g., ColorBrewer, viridis) - Test with colour blindness simulators

### Rainbow Colour Map Problems

- **Not perceptually uniform** -- some bands wider, others narrower

- Creates **false boundaries** where hue changes abruptly
- Obscures data structure
- Preferred alternative: **perceptually uniform maps** (viridis, inferno, magma)

**Exam Tip:** Know why the rainbow colour map is problematic and what to use instead. This is a common question on IV exams.

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## 9. Depicting Quantitative Data

### Fundamental Chart Types

| Chart               | Data Type                       | Strength                               | Weakness                               |
|---------------------|---------------------------------|----------------------------------------|----------------------------------------|
| <b>Bar chart</b>    | Categorical x Quantitative      | Precise comparison via aligned lengths | Only 1-2 variables                     |
| <b>Scatter plot</b> | Quantitative x Quantitative     | Shows correlation, clusters, outliers  | Overplotting with large N              |
| <b>Line chart</b>   | Temporal/Ordered x Quantitative | Shows trends over time                 | Implies continuity                     |
| <b>Pie chart</b>    | Parts of a whole                | Intuitive for proportions              | Hard to compare slices; limit to ~5    |
| <b>Histogram</b>    | Distribution of one variable    | Shows shape of distribution            | Bin size choice affects interpretation |
| <b>Box plot</b>     | Distribution summary            | Shows median, quartiles, outliers      | Hides distribution shape (bimodality)  |

### Tufte's Principles

- **Data-ink ratio:** Maximise the share of ink devoted to data; erase non-data ink
- $\text{Data-ink ratio} = \text{ink used for data} / \text{total ink used}$
- **Chartjunk:** Unnecessary visual elements -- 3D effects, heavy gridlines, decorative images
- **Lie factor:** The size of an effect shown in the graphic divided by the size of effect in the data
- $\text{Lie factor} = (\text{size of effect in graphic}) / (\text{size of effect in data})$
- Should be close to 1.0; values far from 1.0 are misleading
- **Small multiples:** Series of similar charts, each showing a different subset/condition; enables comparison

### Cleveland and McGill's Perceptual Accuracy Hierarchy

Most accurate to least accurate for quantitative judgements: 1. Position along a common scale 2. Position on identical but non-aligned scales 3. Length 4. Direction / Angle / Slope 5. Area 6. Volume / Curvature 7. Shading / Colour saturation

**Exam Tip:** This hierarchy explains WHY bar charts outperform pie charts. Bar charts use position on a common scale (rank 1); pie charts use angle (rank 4).

## 10. Depicting Multidimensional Data

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### The Challenge

Data with many attributes (dimensions) cannot be directly mapped to 2D screen positions. Techniques aim to show relationships across many dimensions simultaneously.

### Techniques

**Parallel Coordinates** - Each dimension = one vertical axis; axes arranged side by side - Each data item = a polyline connecting its values across axes - **Strengths:** Can display many dimensions; reveals correlations between adjacent axes - **Weaknesses:** Axis ordering matters greatly; cluttered with many items; only adjacent-axis correlations visible - **Interaction:** Brushing (select range on one axis, highlight corresponding lines)

**Scatterplot Matrix (SPLOM)** - Grid of all pairwise scatter plots - Diagonal often shows variable name or histogram - **Strengths:** Shows all pairwise correlations - **Weaknesses:** Scales poorly with dimensions ( $n$  dimensions =  $n(n-1)/2$  plots); small individual plots

**Star Plots / Radar Charts** - Radial axes from centre; data item = polygon connecting values - Each dimension is an axis radiating from the centre - **Strengths:** Good for comparing profiles of individual items - **Weaknesses:** Area distorted by axis ordering; hard to compare across many items

**Chernoff Faces** - Map data dimensions to facial features (eye size, mouth curve, nose length, etc.) - **Strengths:** Exploits human face-recognition ability - **Weaknesses:** Arbitrary mapping; some features more salient than others; not precise

**Heatmaps** - Matrix with colour encoding; rows and columns represent different dimensions - **Strengths:** Compact; good for showing patterns in large matrices - **Weaknesses:** Requires meaningful ordering of rows/columns; colour perception limits

**Dimensionality Reduction** - **PCA (Principal Component Analysis):** Linear projection to lower dimensions; preserves variance - **t-SNE:** Non-linear; preserves local neighbourhood structure - **MDS (Multidimensional Scaling):** Preserves pairwise distances - Project high-dimensional data into 2D for scatter plot display - **Trade-off:** Gain overview but lose individual dimension interpretability

**Glyphs** - Small visual objects where multiple visual variables encode multiple data dimensions - Example: A star glyph at each map location encoding 6 climate variables

| Technique                | Max Practical Dimensions | Interaction Needed?        | Best For                       |
|--------------------------|--------------------------|----------------------------|--------------------------------|
| Parallel Coordinates     | 10-20                    | Yes (brushing, reordering) | Finding clusters, correlations |
| SPLOM                    | 8-12                     | Brushing + linking         | All pairwise relationships     |
| Star Plot                | 5-10                     | Minimal                    | Comparing individual profiles  |
| Chernoff Faces           | 10-15                    | No                         | Informal pattern detection     |
| Heatmap                  | 50+                      | Reordering, zooming        | Large matrices                 |
| Dimensionality Reduction | 100+                     | Limited                    | Overview of high-D data        |

**Exam Tip:** Be able to describe parallel coordinates and SPLOM in detail, including their strengths, weaknesses, and the role of interaction.

## 11. Cognitive Amplification -- Larkin & Simon (1987)

### Paper: "Why a Diagram is (Sometimes) Worth Ten Thousand Words"

#### Central Argument

Diagrams are not inherently better than text. The advantage depends on **computational efficiency** -- how easily an information-processing system can extract and use the information.

#### Key Concepts

**Informational Equivalence:** - Two representations are **informationally equivalent** if all information in one can be derived from the other, and vice versa - A diagram and a set of sentences can contain the SAME information

**Computational Equivalence:** - Even if informationally equivalent, representations may NOT be **computationally equivalent** - Different representations support different operators and search strategies - Computational efficiency is what matters in practice

#### Sentential vs Diagrammatic Representations

| Property             | Sentential                              | Diagrammatic                                                |
|----------------------|-----------------------------------------|-------------------------------------------------------------|
| <b>Organisation</b>  | Sequential (linear)                     | Indexed by location in a plane                              |
| <b>Information</b>   | Explicit in propositions                | Much information implicit in spatial layout                 |
| <b>Search</b>        | May require extensive sequential search | Information grouped by location; often at a single location |
| <b>Inference</b>     | Must compute spatial relationships      | Spatial relationships read off directly                     |
| <b>Adjacent cues</b> | No inherent adjacency                   | Cues to next step often at adjacent location                |

#### Three Advantages of Diagrams

1. **Locality:** Related information is grouped spatially -- reduces search
2. **Minimising labelling:** Location itself carries information -- no need for explicit variable names
3. **Perceptual inference:** Some inferences are made directly by the visual system (e.g., recognising that two lines intersect) rather than requiring symbolic computation

#### The "(Sometimes)" Caveat

Diagrams are NOT always better: - When the problem requires symbolic or algebraic manipulation - When the spatial layout does not map naturally to the problem structure - When the diagram is poorly designed (misleading layout) - When information is best expressed sequentially (e.g., a proof)

#### Implications for Visualisation Design

- **Choose representations that make relevant information explicit** rather than requiring computation
- **Group related information spatially** to reduce search
- **Leverage perceptual inference** -- use visual properties the eye processes automatically
- A good visualisation makes the answer **visible** rather than requiring calculation

**Exam Tip:** The key takeaway from Larkin & Simon is the distinction between informational and computational equivalence. Two representations can contain identical information but differ hugely in how easy they are to USE. This is the theoretical foundation for why visualisation works.

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## 12. Design and Evaluation

### The Design Process

1. **Understand the domain** -- What data? What tasks? What questions?
2. **Abstract the data** -- Identify data types and attributes
3. **Design the encoding** -- Choose marks and channels; apply effectiveness principles
4. **Implement** -- Build the visualisation
5. **Evaluate** -- Does it work? For whom?

### Munzner's Nested Model of Visualisation Design

Four levels, each with its own threats to validity:

| Level                                | Question                             | Threat               |
|--------------------------------------|--------------------------------------|----------------------|
| <b>Domain characterisation</b>       | What do users need?                  | Wrong problem        |
| <b>Data/task abstraction</b>         | How to map domain to abstract types? | Wrong abstraction    |
| <b>Visual encoding / interaction</b> | How to encode and interact?          | Ineffective encoding |
| <b>Algorithm</b>                     | How to compute efficiently?          | Slow algorithm       |

### Evaluation Methods

**Quantitative (Empirical / Controlled Experiments)** - Measure task completion time, error rate, accuracy - Compare Technique A vs Technique B - Statistical analysis (t-tests, ANOVA) - **Strengths:** Objective, reproducible, statistically rigorous - **Weaknesses:** Lab setting may not reflect real use; limited task scope

**Qualitative (Observational / Exploratory)** - Think-aloud protocols, interviews, observation - Understanding HOW and WHY users behave - **Strengths:** Rich insights, discovers unexpected issues - **Weaknesses:** Subjective, hard to generalise, small sample sizes

**Heuristic Evaluation** - Experts evaluate against established principles/guidelines - Cheaper and faster than user studies - Can catch many usability issues early

**Case Studies / Deployment** - Deploy with real users doing real tasks over extended time - Richest ecological validity - Hardest to control and generalise

### Evaluation Metrics

| Metric                  | What It Measures                |
|-------------------------|---------------------------------|
| Task completion time    | Efficiency                      |
| Error rate              | Accuracy                        |
| Insight count           | Discovery effectiveness         |
| Subjective satisfaction | User preference (Likert scales) |
| Learning curve          | Learnability                    |

**Exam Tip:** Be able to explain the difference between quantitative and qualitative evaluation, give an example of each, and state when each is appropriate.

## 13. Network and Graph Visualisation

### Terminology

- **Node (vertex):** An entity
- **Edge (link):** A relationship between two nodes
- **Directed graph:** Edges have direction (arrows)
- **Undirected graph:** Edges are symmetric
- **Weighted graph:** Edges have numeric values
- **Tree:** A connected graph with no cycles (hierarchical)

### Layout Algorithms

**Force-Directed (Spring-Embedder)** - Nodes repel each other (like electric charges) - Edges act as springs pulling connected nodes together - Iterative until equilibrium - **Strengths:** Aesthetically pleasing; clusters emerge naturally - **Weaknesses:** Slow for large graphs; non-deterministic; can get stuck in local optima

**Hierarchical / Layered (Sugiyama)** - Assign nodes to layers; minimise edge crossings - Good for DAGs (directed acyclic graphs) - Shows flow/hierarchy clearly

**Circular** - Nodes arranged on a circle; edges drawn inside - Good for showing connections; can highlight clusters

**Matrix Representation** - Adjacency matrix: rows and columns = nodes; cell colour = edge weight - No edge crossings; good for dense graphs - Hard to follow paths; requires meaningful ordering

### Tree Visualisation Techniques

| Technique                | Description                              | Best For                                      |
|--------------------------|------------------------------------------|-----------------------------------------------|
| <b>Node-link diagram</b> | Traditional tree layout                  | Small-medium trees                            |
| <b>Treemap</b>           | Nested rectangles; area = size attribute | Large hierarchies with quantitative attribute |
| <b>Sunburst</b>          | Radial treemap; rings = levels           | Hierarchical navigation                       |
| <b>Icicle plot</b>       | Rectangular, top-down partition          | Hierarchies with clear levels                 |

### Scalability Challenges

- **Node overplotting:** Too many nodes overlap
- **Edge crossing:** Dense graphs become "hairball"
- **Solutions:** Aggregation, edge bundling, filtering, fisheye distortion, semantic zoom

**Exam Tip:** Be able to compare node-link vs matrix representations for graphs, stating when each is appropriate (node-link for sparse graphs with path-tracing tasks; matrix for dense graphs with connection-checking tasks).

## 14. Temporal Data Visualisation

### Characteristics of Temporal Data

- **Time is special:** It is ordered, continuous, and has natural hierarchical structure (seconds > minutes > hours > days)
- **Cyclic patterns:** Daily, weekly, seasonal
- **Trends:** Long-term increase or decrease
- **Anomalies:** Unexpected deviations

### Techniques

| Technique                       | Description                                          | Use Case                                    |
|---------------------------------|------------------------------------------------------|---------------------------------------------|
| <b>Line chart</b>               | Position encodes value over time on x-axis           | Single time series; trends                  |
| <b>Stacked area chart</b>       | Multiple series stacked; area = cumulative           | Composition over time                       |
| <b>Horizon chart</b>            | Layered bands; divides range into colour-coded bands | Comparing many time series in compact space |
| <b>Sparklines</b>               | Tiny inline line charts (Tufte)                      | Embedding trends in text/tables             |
| <b>Small multiples</b>          | Same chart repeated for different subsets            | Comparing patterns across categories        |
| <b>ThemeRiver / Streamgraph</b> | Organic stacked area, centred                        | Thematic changes over time                  |
| <b>Calendar heatmap</b>         | Grid layout (day x week) with colour                 | Cyclic patterns (day-of-week effects)       |
| <b>Gantt chart</b>              | Horizontal bars showing duration                     | Project scheduling, intervals               |

### Animation vs Small Multiples

| Aspect                  | Animation                     | Small Multiples          |
|-------------------------|-------------------------------|--------------------------|
| <b>Change detection</b> | Hard (change blindness)       | Easy (side-by-side)      |
| <b>Engagement</b>       | High (dynamic, engaging)      | Lower (static)           |
| <b>Comparison</b>       | Relies on memory              | Direct visual comparison |
| <b>Recommended</b>      | For presentation/storytelling | For analysis/comparison  |

**Exam Tip:** Animation is engaging but poor for analytical comparison due to change blindness. Small multiples are almost always better for analysis.

## 15. Rules of Thumb for Visualisation

### Tufte's Principles (Summary)

1. **Above all else, show the data**
2. Maximise the **data-ink ratio**
3. Erase **non-data ink** and redundant data ink
4. **Revise and edit**
5. Avoid **chartjunk** (moiré patterns, heavy grids, 3D effects)
6. Use **small multiples** for comparisons
7. **Lie factor** should be close to 1.0

### Munzner's Key Principles

1. **No unjustified 3D** -- 3D introduces occlusion and perspective distortion; only use for inherently 3D data
2. **Eyes beat memory** -- Put information side by side rather than requiring users to remember
3. **Expressiveness principle**: A visualisation should express ALL and ONLY the information in the data (no more, no less)
4. **Effectiveness principle**: Of all the correct (expressive) encodings, choose the most effective one for the data type

### Shneiderman's Guidelines

1. **Overview first, zoom and filter, then details-on-demand**
2. Enable rapid, incremental, reversible actions
3. Support dynamic queries with immediate feedback
4. Use tight coupling between views

### Common Design Pitfalls

| Pitfall                   | Problem                           | Fix                                         |
|---------------------------|-----------------------------------|---------------------------------------------|
| Truncated y-axis          | Exaggerates differences           | Start y-axis at zero for bar charts         |
| Pie charts for comparison | Angle perception is poor          | Use bar charts instead                      |
| Rainbow colour map        | Non-uniform; false boundaries     | Use perceptually uniform colour maps        |
| Unnecessary 3D            | Occlusion, perspective distortion | Use 2D unless data is inherently 3D         |
| Dual y-axes               | Misleading correlations           | Use two separate charts                     |
| Overplotting              | Can't see density/distribution    | Use transparency, jittering, or aggregation |

**Exam Tip:** "No unjustified 3D" and "start bar chart y-axis at zero" are classic exam points. Be ready to critique a given visualisation.

## 16. Key Concepts Quick Reference

### Essential Definitions

| Term                             | Definition                                                                                                    |
|----------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Information Visualisation</b> | Computer-supported, interactive, visual representation of abstract data to amplify cognition                  |
| <b>Visual variable</b>           | A property of a graphical mark that can encode data (position, size, shape, hue, value, orientation, texture) |
| <b>Pre-attentive processing</b>  | Visual features detected in <200ms without focused attention                                                  |
| <b>Data-ink ratio</b>            | Proportion of a graphic's ink devoted to non-redundant display of data information                            |
| <b>Chartjunk</b>                 | Visual elements that do not carry data information (decorations, 3D effects)                                  |
| <b>Lie factor</b>                | Ratio of the size of effect in the graphic to the size of effect in the data                                  |
| <b>Informational equivalence</b> | Two representations from which the same information can be derived                                            |
| <b>Computational equivalence</b> | Two representations that require the same effort to extract information                                       |
| <b>Expressiveness</b>            | A visualisation encodes all and only the data                                                                 |
| <b>Effectiveness</b>             | The most suitable encoding for the data type is used                                                          |
| <b>Brushing and linking</b>      | Selecting items in one view highlights them in all linked views                                               |
| <b>Focus + context</b>           | Showing detail in the focus area while maintaining surrounding context                                        |
| <b>Dynamic queries</b>           | Interactive widgets (sliders, toggles) that filter data with immediate visual feedback                        |
| <b>Small multiples</b>           | Repeated similar visualisations showing different subsets of data                                             |

### Key People and Contributions

| Person                                       | Contribution                                                             |
|----------------------------------------------|--------------------------------------------------------------------------|
| <b>Jacques Bertin</b>                        | Semiology of Graphics (1967) -- visual variables framework               |
| <b>Ben Shneiderman</b>                       | Visual Information-Seeking Mantra; Task by Data Type Taxonomy (1996)     |
| <b>Edward Tufte</b>                          | Data-ink ratio, chartjunk, lie factor, small multiples                   |
| <b>Tamara Munzner</b>                        | Nested model; marks and channels; effectiveness ranking                  |
| <b>William Cleveland &amp; Robert McGill</b> | Perceptual accuracy hierarchy for graphical elements                     |
| <b>Jeffrey Heer</b>                          | Generalised selection via query relaxation                               |
| <b>Jill Larkin &amp; Herbert Simon</b>       | Diagrams vs sentences; informational vs computational equivalence (1987) |
| <b>Anne Treisman</b>                         | Feature integration theory; pre-attentive processing (1985)              |
| <b>Colin Ware</b>                            | Information Visualization: Perception for Design                         |

## Key Papers to Know

1. **Bertin (1967)** -- Semiology of Graphics -- visual variables
2. **Shneiderman (1996)** -- "The Eyes Have It" -- 7 data types x 7 tasks; the mantra
3. **Heer et al.** -- Generalized Selection via Interactive Query Relaxation
4. **Larkin & Simon (1987)** -- "Why a Diagram is (Sometimes) Worth Ten Thousand Words"
5. **Treisman (1985)** -- Pre-attentive processing in vision
6. **Cleveland & McGill (1984)** -- Graphical perception; perceptual accuracy ranking
7. **Munzner (2015)** -- Visualization Analysis and Design (textbook used in course)

## Visual Variables Summary Table

| Variable    | Nom | Ord | Quant | Selective | Assoc | Ordered | Quantitative |
|-------------|-----|-----|-------|-----------|-------|---------|--------------|
| Position    | G   | G   | G     | Y         | Y     | Y       | Y            |
| Size        | M   | G   | G     | Y         | N     | Y       | Y            |
| Shape       | G   | P   | P     | N         | Y     | N       | N            |
| Hue         | G   | P   | P     | Y         | Y     | N       | N            |
| Value       | P   | G   | M     | Y         | N     | Y       | N            |
| Orientation | M   | P   | P     | Y         | Y     | N       | N            |
| Texture     | G   | M   | P     | Y         | Y     | N       | N            |

G=Good, M=Marginal, P=Poor, Y=Yes, N=No

## Shneiderman's 7 Data Types (Mnemonic: "1-2-3-T-M-Tr-N")

1D - Linear (text, lists)  
2D - Planar (maps, floor plans)  
3D - Volumetric (molecules, buildings)  
T - Temporal (stock prices, timelines)  
M - Multidimensional (database records)  
Tr - Tree (file systems, org charts)  
N - Network (social networks, web)

## Shneiderman's 7 Tasks (Mnemonic: "OZF-DRHE")

Overview  
Zoom  
Filter  
Details-on-demand  
Relate  
History  
Extract

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## 17. Exam Strategy

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### General Approach (1h 30m)

- **Read all questions first** (5 minutes)
- **Allocate time proportionally** to marks
- **Leave 5 minutes** at the end for review
- If stuck, move on and come back

### How to Answer IV Questions

**For "describe a visualisation" questions:** 1. State the data type(s) and attribute types 2. Choose appropriate marks and channels, justify with effectiveness ranking 3. Describe the interaction (apply Shneiderman's mantra) 4. Mention evaluation -- how would you know it works?

**For "critique a visualisation" questions:** 1. Identify what works (correct encoding choices) 2. Identify problems (wrong visual variable for data type, chartjunk, misleading scales, poor colour) 3. Suggest specific improvements with justification 4. Reference principles (Tufte, Munzner, Bertin)

**For "compare two techniques" questions:** 1. Use a structured comparison table 2. Consider: expressiveness, effectiveness, scalability, tasks supported 3. State when each is preferred

**For theory questions:** 1. Define the concept precisely 2. Give a concrete example 3. Explain WHY it matters for visualisation design 4. Reference the relevant paper/person

### Common Exam Topics (High Priority)

1. Bertin's visual variables and their properties (associative, selective, ordered, quantitative)
2. Shneiderman's mantra and 7 tasks
3. Pre-attentive processing and pop-out
4. Larkin & Simon -- informational vs computational equivalence
5. Channel effectiveness ranking (Cleveland & McGill / Munzner)
6. Colour: why rainbow is bad; colour blindness; sequential vs diverging vs categorical
7. Multidimensional visualisation techniques (parallel coordinates, SPLOM)
8. Data-ink ratio and chartjunk (Tufte)
9. Evaluation methods (qualitative vs quantitative)
10. Network visualisation (node-link vs matrix)

**Final Exam Tip:** When in doubt, connect your answer back to the **Visual Information-Seeking Mantra** and the **effectiveness ranking**. These two frameworks answer most design questions. Always justify visual encoding choices by referencing data types and perceptual properties.

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*These notes cover the full COMPSCI5099 Information Visualisation curriculum based on course materials (Bertin slides, Shneiderman 1996, Heer et al., Larkin & Simon 1987, Summary and Response Document 2026) and standard MSc-level IV content including Munzner's textbook framework. Prepared for closed-book exam revision.*